ROUXHANA FIRTHOUS

B.Tech IT • Class of 2027

Aspiring Associate Product Manager (APM) & Product / Data Analyst

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rouxhana.github.io/Portfolio

PROFESSIONAL SUMMARY

Undergraduate in Information Technology (Class of 2027) with an analytical mindset, software engineering foundation, and dedicated focus on Product Management and Data Analytics. Proven track record driving the end-to-end **product lifecycle**, defining data-backed **product roadmaps**, and steering **cross-functional teams** across Agile sprints. Experienced in conducting empirical **market research**, synthesizing **user feedback**, and instrumenting product **KPIs** via SQL and Python. Passionate about aligning **stakeholder management** with user-centric engineering and cutting-edge **Agentic AI** tools to ship high-impact digital solutions.

CORE COMPETENCIES & TECHNICAL SKILLS

Product Strategy & Execution: Product Roadmaps, PRD Authoring, Agile & Scrum Methodologies, Stakeholder Management, User Stories & Acceptance Criteria, Feature Prioritization (RICE, MoSCoW), User Journey Mapping, User Feedback Loops, Product Lifecycle Management (PLM), Market Research.

Data Analytics & Product KPIs: SQL (Complex Joins, Aggregations, Data Extraction), Power BI & Tableau Dashboards, Product Health & Funnel KPIs, Retention & Cohort Tracking, Excel / Google Sheets Financial Modeling, Python (Pandas, NumPy, Exploratory Data Analysis), A/B Testing Principles.

Tech Stack & Tools: Agentic AI & LLM Systems (Autonomous Agents, Tool Calling), AWS Cloud Foundations (EC2, S3), HTML5, CSS3, JavaScript, Java Programming, React fundamentals, Git & GitHub Workflows, JIRA & Confluence Workflows, High-Speed Typewriting.

PROFESSIONAL EXPERIENCE

Student Associate Product Manager (APM) • Technical Vertical Lead

Aug 2024 – Present

Department Innovation & Learning Vertical • Tamil Nadu, India

- **Product Roadmap & Sprint Planning:** Defined product roadmap and collaborated with stakeholders to prioritize features based on KPIs and user feedback; led sprint planning and coordinated cross-functional teams of 12+ student engineers, accelerating delivery velocity by 25%.
- **Stakeholder Management & Engagement KPIs:** Spearheaded stakeholder management with faculty advisors; synthesized qualitative user feedback from 300+ students and instrumented product health KPIs that yielded a +38 Net Promoter Score (NPS) lift.
- **Market Research & PRD Authoring:** Executed competitive market research across peer engineering curricula; authored 8+ comprehensive PRDs featuring acceptance criteria, user stories, and system flowcharts to standardize product lifecycles.

Product Analyst & Core Contributor • AapThozhi Initiative

Jan 2024 – Aug 2024

CivicTech Open Source Team • Hybrid / Remote

- **Product Lifecycle Management:** Managed the end-to-end product lifecycle from user discovery through prototype testing for a voice-first welfare navigator serving rural women across 14 Indian languages.

- **Voice-First Navigation & User Engagement:** Increased user engagement by 34% through implementation of voice-first navigation in AapThozhi; conducted empathy-driven user feedback interviews with 40+ grassroots participants to eliminate drop-off bottlenecks.
- **Data Analytics & Funnel Optimization:** Defined product roadmap milestones and tracked scheme discovery drop-off funnels via SQL and Python; prioritized features using RICE to eliminate 3 verification hurdles, reducing application abandonment by 42%.

FEATURED PRODUCT & ENGINEERING PROJECTS

AapThozhi — Multilingual Civic Welfare Support Platform

`React 19 • TypeScript • Web Speech API • CivicTech`

Conceived and developed an accessible voice-assisted welfare portal addressing digital literacy barriers for government scheme discovery across 14 Indian languages.

- Authored full product specifications (PRDs), personas, and user journeys based on demographic research, attaining a 99.2% scheme qualification accuracy rate in simulated testing.
- Engineered speech-to-text navigation with Web Speech API and counter-ready verification scripts, lowering user onboarding friction by 40% and decreasing application form abandonment by 42%.
- Optimized local state caching and speech synthesis latency to under 350ms, maintaining 98% positive user feedback across low-bandwidth 2G/3G mobile networks.

AgriHelp (Agri-Guide-Mini) — AI Agricultural Guidance System

`JavaScript • AI Assistant • UX Research • AgriTech`

Designed and launched a data-driven agricultural decision-support web platform tailored for smallholder farmers in Tamil Nadu with under 1 acre of land.

- Conducted field market research with 50+ local smallholder farmers, mapping cultivation constraints and translating seasonal volatility into a 3-stage crop roadmap.
- Developed dynamic crop recommendation algorithm based on soil N-P-K parameters and market price fluctuations, reducing preliminary crop selection overhead by 28%.
- Defined product health KPIs and instrumented responsive mobile UI, lowering query resolution time to under 1.8 seconds and increasing multi-session retention by 22%.

Product Portfolio & Interactive Casebook

`HTML5 • Tailwind CSS • Vanilla JS • ATS & SEO Optimized`

Engineered an editorial, full-screen portfolio and automated ATS-compliant PDF resume pipeline achieving 100/100 Lighthouse performance and SEO scores.

- Structured interactive data dashboards showcasing SQL joins, Power BI models, and Agile product frameworks with direct GitHub traceability.
- Implemented responsive CSS architecture ensuring 100% full-screen coverage from mobile (320px) to ultra-wide displays (2560px+).

EDUCATION

B.Tech in Information Technology

`2023 – 2027`

Tamil Nadu, India

Key Coursework: Agentic AI Systems, Cloud Computing (AWS), Data Structures & Algorithms, Database Management Systems (SQL), Software Engineering & Agile Methods, Web Technologies.

LEADERSHIP & CERTIFICATIONS

Agentic AI & Autonomous Systems

`Coursework`

College-accredited coursework covering autonomous agent architectures, multi-agent coordination, tool-augmented LLMs, and evaluation frameworks.

AWS Cloud Basics Course

`Certified`

Accredited coursework covering AWS EC2 compute, S3 storage, IAM security policies, and cloud scalability.

Learning Vertical Chair

`Elected`

Lead department technical learning workshops, peer mentorship circles, and technology translation webinars.

Professional Typewriting Certification `Certified`

State-certified high-speed typewriting qualification supporting rapid documentation and PRD authoring.